SUPPORTING INFORMATION

A Theoretical Study of Formamide Decomposition Pathways

Vinh Son Nguyen,[1] Heather L. Abbott,[2,*] M. Michele Dawley,[2] Thomas M. Orlando,[2] Jerzy Leszczynski[3] and Minh Tho Nguyen[1,*]

[1]*Department of Chemistry, and LMCC-Mathematical Modeling and Computational Science Center, Katholieke Universiteit Leuven, B-3001 Leuven, Belgium*
[2]*School of Chemistry and Biochemistry, Georgia Institute of Technology, Atlanta, GA 30332-0400.*
[3]*Interdisciplinary Center for Nanotoxicity, Department of Chemistry, Jackson State University, Jackson, MS 39217-0510*

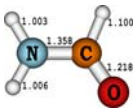
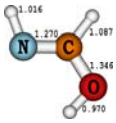
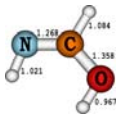
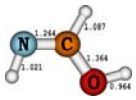
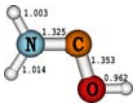
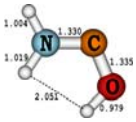
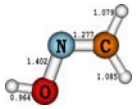
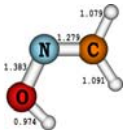
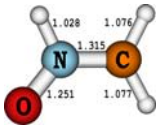
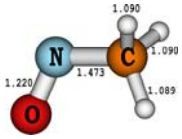
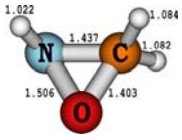
Supporting Information Available

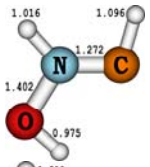
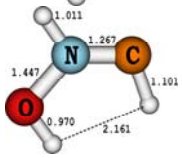
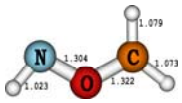
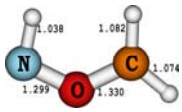
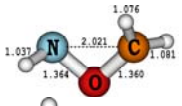
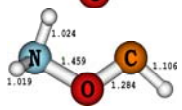
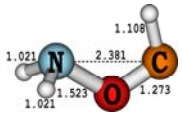
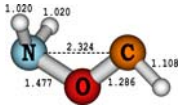
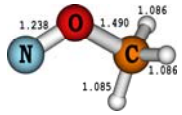
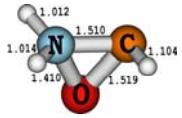
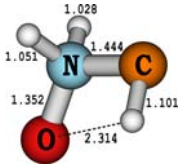
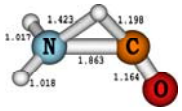
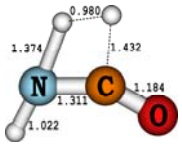
The Supporting Information for this document includes tables with structures, rotational constants, vibrational frequencies and total energies for all optimized geometries. In addition, the entropy, zero point correction and thermal correction for formamide, reactive intermediates and transition states are available. Additional plots illustrating the pressure dependence of RRKM rate constants, k_i, are also provided. This information is available free of charge via the Internet at http://pubs.acs.org.

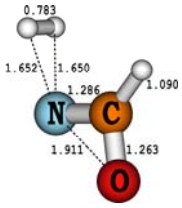
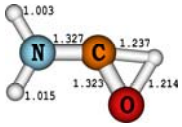
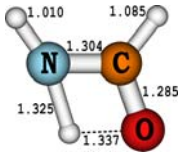
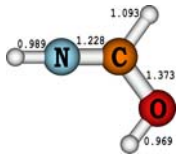
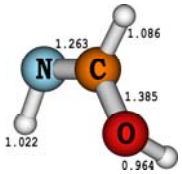
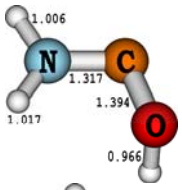
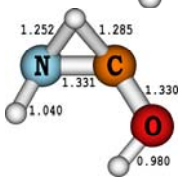
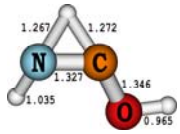
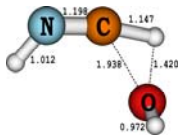
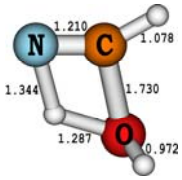
* Emails: heather.abbott-lyon@gatech.edu, minh.nguyen@chem.kuleuven.be

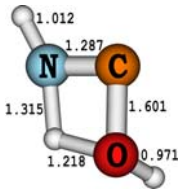
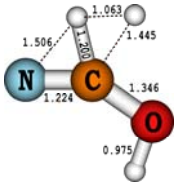
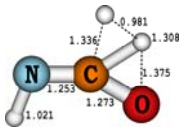
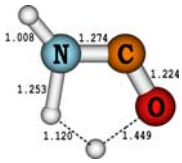
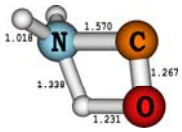
Table S1
Vibrational frequencies (at 298 K), rotational constants (in GHZ) of Formamide, intermediates and the transition states involved in the decomposition of Formamide.

N—C—O PES's

Species	Structures [a]	Rotational Constants, GHZ			Vibrational frequencies, cm^{-1} [b]
NH_2CHO		72.03840	11.17477	9.67410	174, 557, 635, 1031, 1048, 1274, 1405, 1618, 1764, 3036, 3605, 3764
INT-1a		69.30074	11.30740	9.72124	580, 618, 829, 1031, 1065, 1185, 1367, 1379, 1692, 3179, 3533, 3725
INT-1b		66.27432	11.05377	9.47368	516, 577, 835, 1062, 1069, 1119, 1341, 1402, 1689, 3232, 3458, 3764
INT-1c		69.25678	11.05905	9.53628	401, 589, 849, 1051, 1054, 1167, 1298, 1409, 1715, 3182, 3469, 3808
INT-2a		69.17476	11.56671	9.90971	555, 556, 630, 772, 1022, 1180, 1277, 1426, 1647, 3497, 3735, 3840
INT-2b		65.33272	11.58440	9.83969	512, 584, 592, 751, 1042, 1156, 1310, 1392, 1650, 3424, 3530, 3711
INT-3a		65.96869	11.76537	9.98463	419, 527, 793, 898, 961, 1166, 1338, 1444, 1659, 3148, 3298, 3805
INT-3b		63.72115	11.83265	9.97952	438, 555, 770, 919, 931, 1154, 1370, 1449, 1654, 3099, 3285, 3627
INT-4		74.55671	11.59181	10.03206	567, 653, 744, 966, 1078, 1297, 1473, 1492, 1657, 3215, 3336, 3370
INT-5		59.38191	11.38263	10.16508	156, 574, 893, 975, 1155, 1366, 1447, 1451, 1528, 3056 , 3166, 3187
INT-6		25.98172	14.25198	23.74782	715, 897, 970, 1076, 1210, 1230, 1249, 1347, 1530, 3134, 3247, 3423

INT-7a		68.50779	11.52273	9.86370	455, 560, 676, 899, 1008, 1177, 1362, 1455, 1584, 3076, 3497, 3655
INT-7b		63.85893	11.05539	9.42390	259, 487, 679, 750, 1055, 1120, 1287, 1470, 1581, 3033, 3597, 3724
INT-8a		78.64970	10.16618	11.67532	299, 533, 602, 787, 1050, 1166, 1442, 1515, 1858, 3199, 3388, 3413
INT-8b		72.58361	11.73494	10.10175	263, 532, 550, 804, 1015, 1165, 1445, 1507, 1638, 3175, 3217, 3375
INT-8c		78.72175	11.66984	10.16323	299, 532, 603, 787, 1049, 1166, 1442, 1516, 1587, 3198, 3388, 3411
INT-9a		64.18492	11.69109	10.17484	257, 559, 770, 836, 1221, 1308, 1368, 1408, 1585, 2963, 3379, 3522
INT-9b		58.74426	11.11359	9.82654	122, 459, 612, 820, 1157, 1234, 1272, 1417, 1578, 2932, 3410, 3520
INT-9c		64.18481	11.69110	10.17484	257, 559, 770, 836, 1221, 1308, 1368, 1408, 1585, 2963, 3379, 3522
INT-10		59.75664	11.33057	10.15786	127, 556, 754, 1059, 1150, 1344, 1447, 1464, 1468, 3090, 3228, 3241
INT-11		25.78450	23.28981	13.97716	597, 811, 850, 1009, 1110, 1152, 1209, 1338, 1623, 2953, 3479, 3622
INT-12		25.77651	23.29039	13.97500	596, 811, 850, 1009, 1110, 1152, 1209, 1338, 1623, 2953, 3479, 3623
NH_3-ts		72.46230	7.82978	7.25908	**-1704**, 208, 251, 397, 579, 735, 1184, 1440, 1796, 2356, 3421, 3589
H_2-ts1		78.60184	10.56600	9.31398	**-1951**, 319, 612, 693, 821, 1208, 1180, 1300, 1694, 2016, 2206, 3455

H_2-ts2		50.92250	13.01883	10.50487	**-1109**, 268, 456, 480, 652, 684, 963, 1190, 1455, 1594, 3155, 3840
tsH-1,2		75.44539	10.99961	9.59998	**-1910**, 398, 456, 523, 675, 1038, 1312, 1385, 1612, 2613, 3501, 3736
tsH-1,3		57.48115	13.38523	10.85703	**-1883**, 446, 848, 1012, 1088, 1137, 1221, 1422, 1609, 2108, 3227, 3598
tsR-1a1b		72.20877	10.85920	9.43961	**-1057**, 351, 599, 623, 993, 1039, 1256, 1415, 1799, 3110, 3737, 3917
tsR-1b1c		64.35402	10.86992	9.56747	**-473**, 574, 805, 1014, 1076, 1179, 1239, 1395, 1672, 3198, 3462, 3811
tsR-2a2b		61.69276	11.41007	9.91435	**-666**, 557, 687, 761, 947, 1087, 1205, 1371, 1637, 3454, 3692, 3785
tsH-1b2b		65.40582	11.25408	9.80052	**-1866**, 498, 601, 648, 947, 1102, 1171, 1344, 1470, 2407, 3170, 3528
tsH-1c2a		68.80235	11.17561	9.82796	**-1887**, 487, 600, 627, 1000, 1086, 1175, 1312, 1498, 2410, 3254, 3797
INT-1b-tsH_2O		64.91814	7.59934	6.95469	**-1464**, 247, 294, 607, 640, 735, 769, 1085, 1811, 2876, 3582, 3721
INT-1c-tsH_2O		45.85582	10.97155	9.08258	**-1978**, 490, 554, 624, 767, 871, 1070, 1271, 1762, 1839, 3295, 3715

INT-2a-tsH_2O		49.53216	12.20604	10.03990	**-1662**, 515, 576, 701, 822, 973, 1141, 1289, 1542, 1929, 3587, 3733
INT-1a-tsH_2		73.54123	10.50084	9.18879	**-2003**, 360, 512, 566, 789, 974, 1046, 1275, 1560, 2014, 2524, 3669
INT-1c-tsH_2		81.37734	10.57713	9.36048	**-2011**, 518, 583, 685, 869, 966, 1062, 1332, 1808, 2099, 2301, 3490
INT-2b-tsH_2		67.19267	12.54452	10.57097	**-1712**, 263, 689, 780, 937, 1115, 1149, 1387, 1447, 1776, 1899, 3669
INT-2b-tsNH_3		50.31962	12.24370	10.41182	**-1659**, 331, 655, 850, 921, 1003, 1349, 1432, 1540, 1917, 3448, 3569

[a]Geometry optimization is performed at MP2/aug-cc-pVTZ level.

[b]Vibrational frequencies (non scaled) are obtained from a vibrational analysis of the geometry optimized structures for these species obtained from Moller Plesset perturbation calculations performed at the MP2/aug-cc-pVDZ level. Frequencies indicated in boldface correspond to imaginary frequencies of transition states.

Table S1 (continued)

C—N—O PES's

tsR-11	29.86850	19.22350	13.08759	**-1375**, 655, 789, 987, 1070, 1105, 1238, 1268, 1574, 2956, 3359, 3588
tsC-119a	36.96485	15.02841	11.82341	**-730**, 387, 771, 796, 1043, 1128, 1248, 1404, 1590, 2875, 3483, 3627
tsH-116	27.70778	19.98831	13.02938	**-971**, 550, 761, 963, 1075, 1115, 1230, 1347, 1383, 2587, 3089, 3445
HtsH-111a	24.72930	21.98616	13.12411	**-1579**, 558, 771, 936, 957, 1086, 1099, 1323, 1354, 2443, 2969, 3420
tsH-111b				

[a]Geometry optimization is performed at MP2/aug-cc-pVTZ level.

[b]Vibrational frequencies (non scaled) are obtained from a vibrational analysis of the geometry optimized structures for these species obtained from Moller Plesset perturbation calculations performed at the MP2/aug-cc-pVDZ level. Frequencies indicated in boldface correspond to imaginary frequencies of transition states.

Table S2
Processes and Energies in **N—C—O PES's**. Energy values (kcal/mol) were obtained at CCSD(T)/CBS level

One-Step Decomposition

Lowest energy pathway to H_2 + HNCO ***(k_1)***
Formamide (**0.0**) → H_2-ts1 (**78.8**) → H_2 + HNCO (**15.7**)

Alternative pathway to H_2 + HNCO
Formamide (**0.0**) → H_2-ts2 (**112.5**) → H_2 + HNCO (**15.7**)

Pathway to give NH_3 + CO ***(k_2)***
Formamide (**0.0**) → NH_3-ts (**80.5**) → CO + NH_3 (**6.2**)

Multi-Step Decomposition

Alternative pathway to NH_3 + CO ***(k_3)***
Formamide (**0.0**) → tsH-1,2 (**72.2**) → INT-2a (**36.8**) → tsR-2a2b (**51.4**) → INT-2b (**42.4**) → INT-2b-tsNH_3 (**77.2**) → NH_3 + CO (**6.2**)

Lowest energy pathway to H_2O + HNC ***(k_4)***
Formamide (**0.0**) → tsH-1,3 (**43.9**) → INT-1a (**11.0**) → tsR-1a1b (**34.0**) → INT-1b (**14.3**) → INT-1b-tsH_2O (**82.7**) → H_2O + HNC (**31.2**)

Lowest energy pathway to H_2O + HCN ***(k_5)***
Formamide (**0.0**) → tsH-1,3 (**43.9**) → INT-1a (**11.0**) → tsR-1a1b (**34.0**) → INT-1b (**14.3**) → tsR-1b1c (**19.7**) → INT-1c (**14.5**) → INT-1c-tsH_2O (**73.6**) → H_2O + HCN (**16.9**)

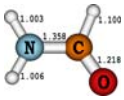
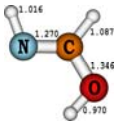
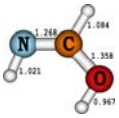
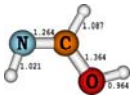
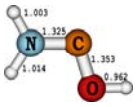
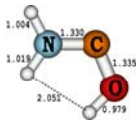
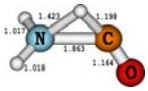
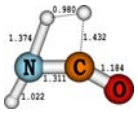
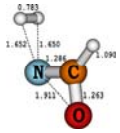
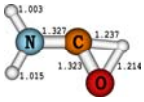
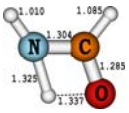
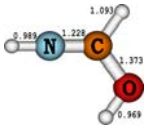
Alternative pathway to H_2O + HNC
Formamide (**0.0**) → tsH-1,2 (**72.2**) → INT-2a (**36.8**) → INT-2a-tsH_2O (**83.7**) → H_2O + HNC (**31.2**)

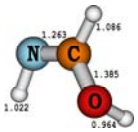
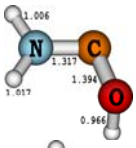
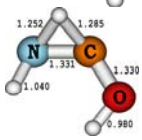
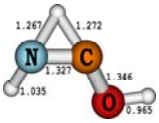
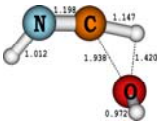
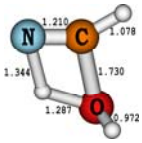
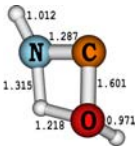
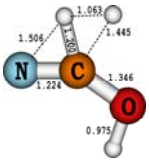
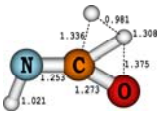
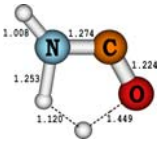
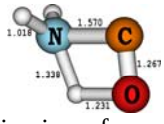
Pathway to H_2 + HOCN
Formamide (**0.0**) → tsH-1,3 (**43.9**) → INT-1a (**11.0**) → INT-1a-tsH_2 (**112.5**) → H_2 + HOCN (**40.3**)

Alternative pathways to H_2 + HNCO
Formamide (**0.0**) → tsH-1,3 (**43.9**) → INT-1a (**11.0**) → tsR-1a1b (**34.0**) → INT-1b (**14.3**) → tsR-1b1c (**19.7**) → INT-1c (**14.5**) → INT-1c-tsH_2 (**91.5**) → H_2 + HNCO (**15.7**)

Formamide (**0.0**) → tsH-1,2 (**72.2**) → INT-2a (**36.8**) → tsR-2a2b (**51.4**) → INT-2b (**42.4**) → INT-2b-tsH_2 (**81.3**) → H_2 + HNCO (**15.7**)

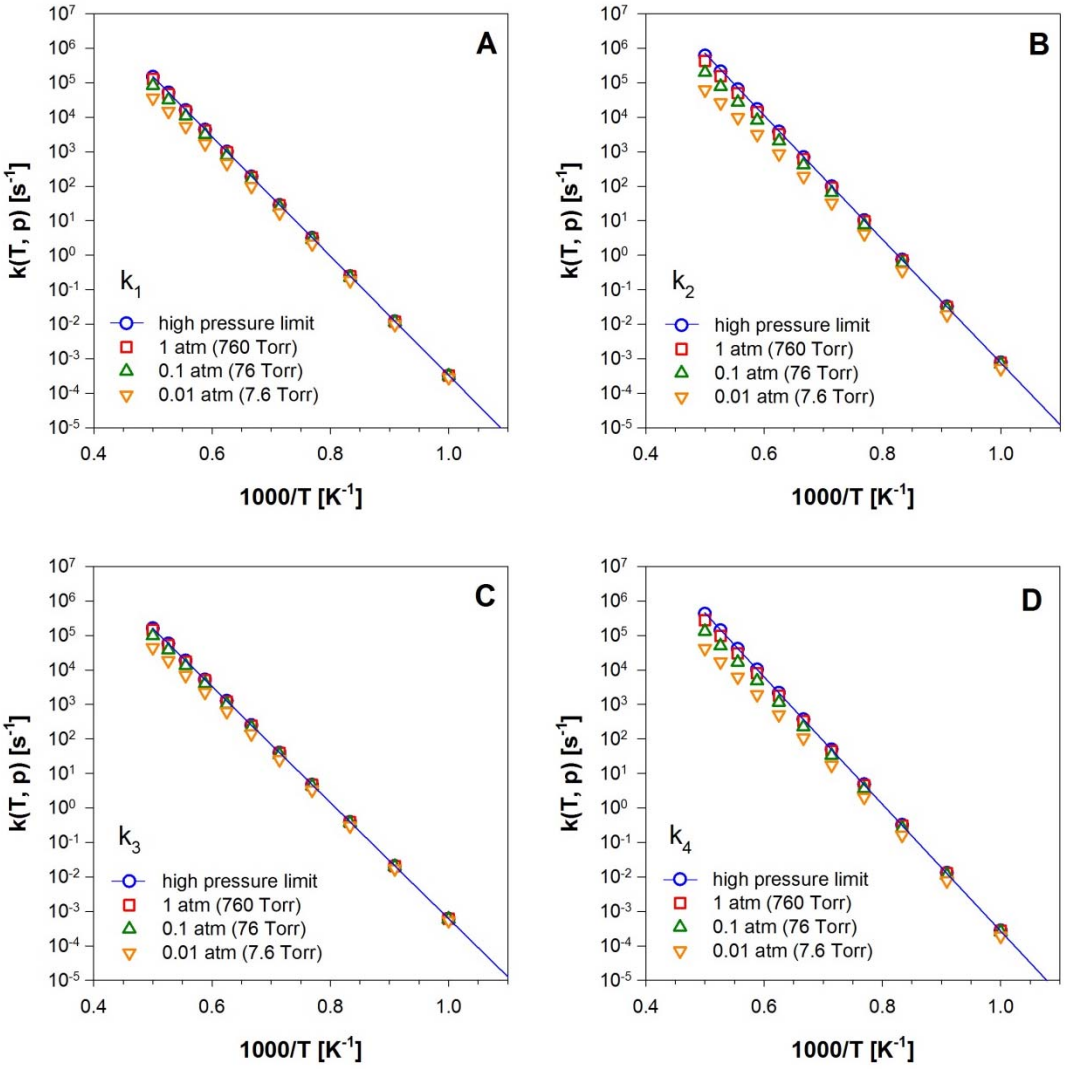
Table S3. Zero-point energies (ZPE in kcal/mol, MP2/aug-cc-pVDZ), thermal corrections (H_{298}-H_0 in kcal/mol), and entropies (S, cal/K.mol) of formamide, intermediates and the transition states involved in the decomposition of formamide.

Species	Structures [a]	S	ZPE	H_{298}-H_0
NH_2CHO		62.1	28.5	3.0
INT-1a		59.9	28.9	2.7
INT-1b		60.2	28.7	2.7
INT-1c		60.6	28.5	2.8
INT-2a		60.5	28.8	2.8
INT-2b		60.6	28.1	2.8
NH_3-ts		64.8	22.8	3.4
H_2-ts1		61.0	21.9	2.9
H_2-ts2		61.4	21.9	2.9
tsH-1,2		61.4	24.7	3.0
tsH-1,3		59.8	25.3	2.6
tsR-1a1b		61.1	26.9	2.9

tsR-1b1c		59.7	27.8	2.6
tsR-2a2b		60.0	27.4	2.7
tsH-1b2b		60.4	24.1	2.8
tsH-1c2a		60.4	24.7	2.8
INT-1b-tsH_2O		64.0	23.1	3.3
INT-1c-tsH_2O		61.2	23.2	2.8
INT-2a-tsH_2O		60.6	24.0	2.8
INT-1a-tsH_2		62.7	21.1	3.2
INT-1c-tsH_2		60.3	22.5	2.8
INT-2b-tsH_2		60.9	21.6	2.9
INT-2b-tsNH_3		60.8	24.3	2.8

[a]Geometry optimization is performed at MP2/aug-cc-pVTZ level.

Figure S1. Variations of apparent rates with pressure and temperature.



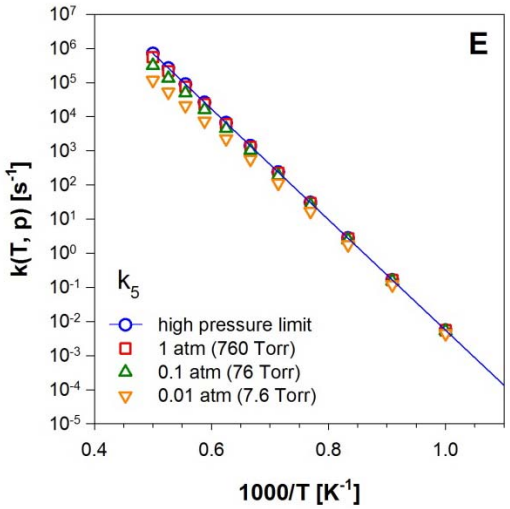


Figure S2. Variations of apparent rates with pressure in the lower temperature region (i.e., T = 200-800 K). The ratio of rate constants does not vary with pressure (i.e., $k_5 > k_3 > k_1 > k_2 > k_4$); however, the absolute value of the rate constants does vary with pressure.

Supporting Information

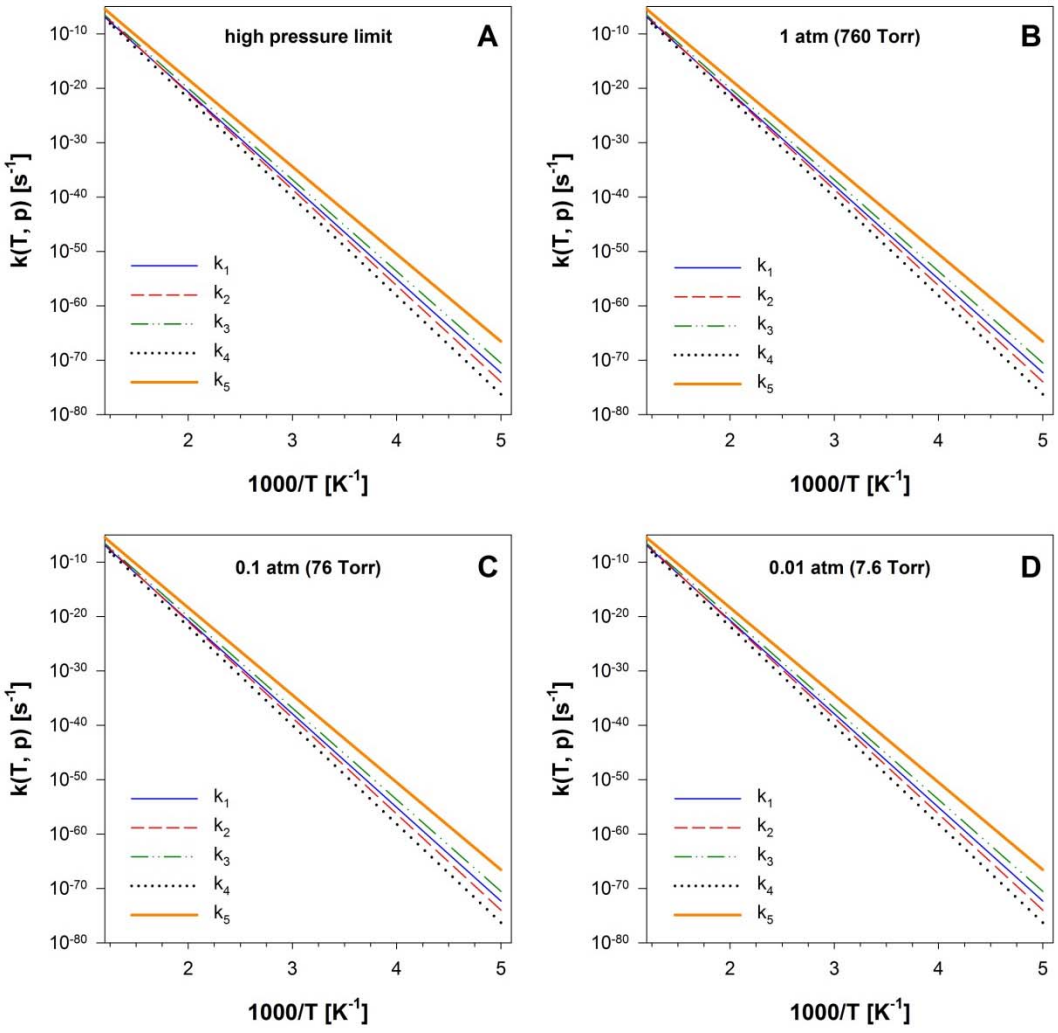


Figure S3. Variations of apparent rates with pressure in the upper temperature region (i.e., T = 800-2000 K). The ratio of rate constants is consistent for all but the lowest pressure region (i.e., $k_5 > k_2 > k_4 > k_3 > k_1$ except for 0.01 atm when $k_5 > k_2 > k_3 > k_4 > k_1$).

